

11.2.4 Implicitly shifted bidiagonal QR algorithm

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Converting a (tridiagonal) implicitly shifted QR algorithm into a (bidiagonal) implicitly shifted QR algorithm now hinges on some key insights, which we will illustrate with a 4×4 example.

- We start with a bidiagonal matrix $B^{(k)}$

$$B^{(k)} = \begin{pmatrix} \beta_{0,0} & \beta_{0,1} & 0 & 0 \\ 0 & \beta_{1,1} & \beta_{1,2} & 0 \\ 0 & 0 & \beta_{2,2} & \beta_{2,3} \\ 0 & 0 & 0 & \beta_{3,3} \end{pmatrix},$$

which is our "current iteration."

- If we explicitly form $T^{(k)} = B^{(k)T} B^{(k)}$, then we would have to form

$$\begin{aligned} T^{(k)} &= \begin{pmatrix} \tau_{0,0} & \tau_{1,0} & 0 & 0 \\ \tau_{1,0} & \tau_{1,1} & \tau_{2,1} & 0 \\ 0 & \tau_{2,1} & \tau_{2,2} & \tau_{3,2} \\ 0 & 0 & \tau_{3,2} & \tau_{3,3} \end{pmatrix} \\ &= \begin{pmatrix} \beta_{0,0}^2 & \beta_{0,1}\beta_{0,0} & 0 & 0 \\ \beta_{0,1}\beta_{0,0} & \beta_{0,1}^2 + \beta_{1,1}^2 & \beta_{1,2}\beta_{1,1} & 0 \\ 0 & \beta_{1,2}\beta_{1,1} & \beta_{1,2}^2 + \beta_{2,2}^2 & \beta_{2,3}\beta_{2,2} \\ 0 & 0 & \beta_{2,3}\beta_{2,2} & \beta_{2,3}^2 + \beta_{3,3}^2 \end{pmatrix} \end{aligned}$$

- The Francis Implicit QR Step would then compute a first Givens' rotation so that

$$\underbrace{\begin{pmatrix} \gamma_0 & -\sigma_0 \\ \sigma_0 & \gamma_0 \end{pmatrix}}_{G_0^T} \begin{pmatrix} \tau_{0,0} - \tau_{3,3} & \\ & \tau_{1,0} \end{pmatrix} = \begin{pmatrix} \times & \\ & 0 \end{pmatrix}. \tag{11.2.1}$$

- With this Givens' rotation, it would introduce a bulge

$$\begin{aligned} &\left(\begin{array}{cc|cc} \times & \times & \color{red}{\times} & 0 \\ \times & \times & \times & 0 \\ \color{red}{\times} & \times & \times & \times \\ 0 & 0 & \times & \times \end{array} \right) \\ &= \left(\begin{array}{c|cc} G_0^T & 0 & 0 \\ \hline 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right) \left(\begin{array}{cc|cc} \tau_{0,0} & \tau_{1,0} & 0 & 0 \\ \tau_{1,0} & \tau_{1,1} & \tau_{2,1} & 0 \\ \hline 0 & \tau_{2,1} & \tau_{2,2} & \tau_{3,2} \\ 0 & 0 & \tau_{3,2} & \tau_{3,3} \end{array} \right) \left(\begin{array}{c|cc} G_0 & 0 & 0 \\ \hline 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right) \end{aligned}$$

- Finally, the bulge would be chased out

$$\begin{aligned} T^{(k+1)} &= \begin{pmatrix} \times & \times & 0 & 0 \\ \times & \times & \times & 0 \\ 0 & \times & \times & \times \\ 0 & 0 & \times & \times \end{pmatrix} \\ &= \left(\begin{array}{cc|cc} 1 & 0 & 0 & \\ \hline 0 & 1 & 0 & \\ 0 & 0 & G_2^T & \end{array} \right) \\ &\quad \underbrace{\left[\left(\begin{array}{cc|cc} 1 & 0 & 0 & 0 \\ \hline 0 & G_1^T & 0 & \\ 0 & 0 & 1 & \end{array} \right) \left(\begin{array}{cc|cc} \times & \times & \color{red}{\times} & 0 \\ \times & \times & \times & 0 \\ \color{red}{\times} & \times & \times & \times \\ \hline 0 & 0 & \times & \times \end{array} \right) \left(\begin{array}{cc|cc} 1 & 0 & 0 & 0 \\ \hline 0 & G_1 & 0 & \\ 0 & 0 & 1 & \end{array} \right) \right] \left(\begin{array}{cc|cc} 1 & 0 & 0 & \\ \hline 0 & 1 & 0 & \\ 0 & 0 & G_2 & \end{array} \right)} \\ &\quad \left(\begin{array}{cc|cc} \times & \times & 0 & 0 \\ \hline \times & \times & \times & \color{red}{\times} \\ 0 & \times & \times & \times \\ 0 & \color{red}{\times} & \times & \times \end{array} \right) \end{aligned}$$

Obviously, this extends.

Let us examine what would happen if we instead apply these Givens' rotations to $B^{(k)T} B^{(k)}$. Since $T^{(k)} = B^{(k)T} B^{(k)}$, we find that

$$\begin{aligned}
T^{(k+1)} &= \begin{pmatrix} \times & \times & 0 & 0 \\ \times & \times & \times & 0 \\ 0 & \times & \times & \times \\ 0 & 0 & \times & \times \end{pmatrix} \\
&= \left(\begin{array}{c|cc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \hline 0 & 0 & G_2^T \end{array} \right) \left(\begin{array}{c|cc} 1 & 0 & 0 \\ 0 & G_1^T & 0 \\ \hline 0 & 0 & 1 \end{array} \right) \left(\begin{array}{c|cc} G_0^T & 0 & 0 \\ 0 & 1 & 0 \\ \hline 0 & 0 & 1 \end{array} \right) \left[\begin{pmatrix} \beta_{0,0} & \beta_{0,1} & 0 & 0 \\ 0 & \beta_{1,1} & \beta_{1,2} & 0 \\ 0 & 0 & \beta_{2,2} & \beta_{2,3} \\ 0 & 0 & & \beta_{3,3} \end{pmatrix}^T \right. \\
&\quad \times \left. \begin{pmatrix} \beta_{0,0} & \beta_{0,1} & 0 & 0 \\ 0 & \beta_{1,1} & \beta_{1,2} & 0 \\ 0 & 0 & \beta_{2,2} & \beta_{2,3} \\ 0 & 0 & & \beta_{3,3} \end{pmatrix} \right] \left(\begin{array}{c|cc} G_0 & 0 & 0 \\ 0 & 1 & 0 \\ \hline 0 & 0 & 1 \end{array} \right) \left(\begin{array}{c|cc} 1 & 0 & 0 \\ 0 & G_1 & 0 \\ \hline 0 & 0 & 1 \end{array} \right) \left(\begin{array}{c|cc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \hline 0 & 0 & G_2 \end{array} \right) \\
&= \left[\begin{pmatrix} \beta_{0,0} & \beta_{0,1} & 0 & 0 \\ 0 & \beta_{1,1} & \beta_{1,2} & 0 \\ 0 & 0 & \beta_{2,2} & \beta_{2,3} \\ 0 & 0 & & \beta_{3,3} \end{pmatrix} \left(\begin{array}{c|cc} G_0 & 0 & 0 \\ 0 & 1 & 0 \\ \hline 0 & 0 & 1 \end{array} \right) \left(\begin{array}{c|cc} 1 & 0 & 0 \\ 0 & G_1 & 0 \\ \hline 0 & 0 & 1 \end{array} \right) \left(\begin{array}{c|cc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \hline 0 & 0 & G_2 \end{array} \right) \right]^T \\
&\quad \times \left[\begin{pmatrix} \beta_{0,0} & \beta_{0,1} & 0 & 0 \\ 0 & \beta_{1,1} & \beta_{1,2} & 0 \\ 0 & 0 & \beta_{2,2} & \beta_{2,3} \\ 0 & 0 & & \beta_{3,3} \end{pmatrix} \left(\begin{array}{c|cc} G_0 & 0 & 0 \\ 0 & 1 & 0 \\ \hline 0 & 0 & 1 \end{array} \right) \left(\begin{array}{c|cc} 1 & 0 & 0 \\ 0 & G_1 & 0 \\ \hline 0 & 0 & 1 \end{array} \right) \left(\begin{array}{c|cc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \hline 0 & 0 & G_2 \end{array} \right) \right].
\end{aligned}$$

The observation now is that if we can find two sequences of Givens' rotations such that

$$\begin{aligned}
B^{(k+1)} &= \begin{pmatrix} \times & \times & 0 & 0 \\ 0 & \times & \times & 0 \\ 0 & 0 & \times & \times \\ 0 & 0 & 0 & \times \end{pmatrix} \\
&= \left(\begin{array}{c|cc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \hline 0 & 0 & \hat{G}_2^T \end{array} \right) \left(\begin{array}{c|cc} 1 & 0 & 0 \\ 0 & \hat{G}_1^T & 0 \\ \hline 0 & 0 & 1 \end{array} \right) \left(\begin{array}{c|cc} \hat{G}_0^T & 0 & 0 \\ 0 & 1 & 0 \\ \hline 0 & 0 & 1 \end{array} \right) \\
&\quad \times \begin{pmatrix} \beta_{0,0} & \beta_{0,1} & 0 & 0 \\ 0 & \beta_{1,1} & \beta_{1,2} & 0 \\ 0 & 0 & \beta_{2,2} & \beta_{2,3} \\ 0 & 0 & & \beta_{3,3} \end{pmatrix} \\
&\quad \times \underbrace{\left(\begin{array}{c|cc} G_0 & 0 & 0 \\ 0 & 1 & 0 \\ \hline 0 & 0 & 1 \end{array} \right) \left(\begin{array}{c|cc} 1 & 0 & 0 \\ 0 & \hat{G}_1 & 0 \\ \hline 0 & 0 & 1 \end{array} \right) \left(\begin{array}{c|cc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \hline 0 & 0 & \hat{G}_2 \end{array} \right)}_Q
\end{aligned} \tag{11.2.2}$$

then, by the Implicit Q Theorem,

$$\begin{aligned}
&B^{(k+1)T} B^{(k+1)} \\
&= \begin{pmatrix} \times & \times & 0 & 0 \\ 0 & \times & \times & 0 \\ 0 & 0 & \times & \times \\ 0 & 0 & 0 & \times \end{pmatrix}^T \begin{pmatrix} \times & \times & 0 & 0 \\ 0 & \times & \times & 0 \\ 0 & 0 & \times & \times \\ 0 & 0 & 0 & \times \end{pmatrix} \\
&= \begin{pmatrix} \times & \times & 0 & 0 \\ \times & \times & \times & 0 \\ 0 & \times & \times & \times \\ 0 & 0 & \times & \times \end{pmatrix} \\
&= T^{(k+1)} \\
&= Q^T T^{(k)} Q.
\end{aligned}$$

If we iterate in this way, we know that $T^{(k)}$ converge to a diagonal matrix (under mild conditions). This means that the matrices $B^{(k)}$ converge to a diagonal matrix, Σ_B . If we accumulate all Givens' rotations into matrices U_B and V_B , then we end up with the SVD of B :

$$B = U_B \Sigma_B V_B^T,$$

modulo, most likely, a reordering of the diagonal elements of Σ_B and a corresponding reordering of the columns of U_B and V_B .

This leaves us with the question of how to find the two sequences of Givens' rotations mentioned in [\(11.2.2\)](#).

- We know G_0 , which was computed from [\(11.2.1\)](#). Importantly, computing this first Givens' rotation requires only that the elements $\tau_{0,0}$, $\tau_{1,0}$, and $\tau_{m-1,m-1}$ of $T^{(k)}$ to be explicitly formed.
- If we apply it to $B^{(k)}$, we introduce a bulge:

$$\left(\begin{array}{c|cc} \times & \times & 0 & 0 \\ \times & \times & \times & 0 \\ 0 & 0 & \times & \times \\ 0 & 0 & 0 & \times \end{array} \right) = \left(\begin{array}{c|cc} \beta_{0,0} & \beta_{0,1} & 0 & 0 \\ 0 & \beta_{1,1} & \beta_{1,2} & 0 \\ 0 & 0 & \beta_{2,2} & \beta_{2,3} \\ 0 & 0 & & \beta_{3,3} \end{array} \right) \left(\begin{array}{c|cc} G_0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right).$$

- We next compute a Givens' rotation, $\hat{G}_0$, that changes the nonzero that was introduced below the diagonal back into a zero.

$$\left(\begin{array}{c|cc} \times & \times & \times & 0 \\ 0 & \times & \times & 0 \\ 0 & 0 & \times & \times \\ 0 & 0 & 0 & \times \end{array} \right) = \left(\begin{array}{c|cc} \hat{G}_0^T & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right) \left(\begin{array}{c|cc} \times & \times & 0 & 0 \\ \times & \times & \times & 0 \\ 0 & 0 & \times & \times \\ 0 & 0 & 0 & \times \end{array} \right).$$

- This means we now need to chase the bulge that has appeared above the superdiagonal:

$$\left(\begin{array}{c|c|c|c} \times & \times & 0 & 0 \\ 0 & \times & \times & 0 \\ 0 & \times & \times & \times \\ 0 & 0 & 0 & \times \end{array} \right) = \left(\begin{array}{c|c|c|c} \times & \times & \times & 0 \\ 0 & \times & \times & 0 \\ 0 & 0 & \times & \times \\ 0 & 0 & 0 & \times \end{array} \right) \left(\begin{array}{c|c|c} 1 & 0 & 0 \\ 0 & \tilde{G}_1 & 0 \\ 0 & 0 & 1 \end{array} \right)$$

- We continue like this until the bulge is chased out the end of the matrix.

The net result is an ***implicitly shifted bidiagonal QR algorithm*** that is applied directly to the bidiagonal matrix, maintains the bidiagonal form from one iteration to the next, and converges to a diagonal matrix that has the singular values of B on its diagonal. Obviously, deflation can be added to this scheme to further reduce its cost.